

A Direction Finder for Power Quality Disturbances Based Upon Disturbance Power and Energy

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Abstract—While existing techniques are capable of automatically identifying and classifying various types of distribution-level power quality disturbances, they do not provide any information about the locations of the disturbance sources. This paper shows that it is possible to use sampled voltage and current waveforms to determine on which side of a recording device a disturbance originates. This is accomplished by examining the energy flow and peak instantaneous power for both capacitor energizing and voltage sag disturbances. We demonstrate the technique by testing it on both ATP-generated waveforms and actual field disturbance data. In both cases, we are able to accurately determine on which side of a recording device the disturbance originates. If enough recording devices are available in a network, the individual readings can be collectively used to pinpoint the locations of disturbance sources.

I. INTRODUCTION

With the proliferation of sensitive electronic loads such as ASDs and microprocessors, the subject of power quality has generated much interest among electric power engineers. Several monitoring projects, such as the EPRI Distribution Power Quality project, have been carried out in an effort to characterize the overall level of power quality on various electric systems [1]. Much work has also been done to develop systems that automatically identify the major types of power quality disturbances measured on distribution networks, such as capacitor switching, voltage sag, and impulsive transients [2-4].

In the near future, responsible parties may be penalized for causing interruptions in the power supply that result in customer downtime. As a result, the ability to locate disturbance sources will become increasingly important. However, there

is not yet a methodology for pinpointing the locations of these sources. In this paper, we show that it is possible to approximately locate the source of a power quality disturbance in a network using sampled voltage and current waveforms taken by a number of recording devices.

For two common disturbance types (capacitor switching and voltage sag), it is possible to determine whether a disturbance originates in front of or behind a recording device (i.e., in the direction of positive power flow or negative power flow). This information may tell the engineer whether the disturbance originated towards or away from the substation, or inside/outside a customer facility, depending on the location of the recording device. If concurrent readings from several recording devices are available, such information may make it possible to further pinpoint the location of the disturbance sources.

II. DISTURBANCE POWER AND ENERGY

Since nonlinear loads can be thought of as sources of power at harmonic frequencies, they can be located by noting that harmonic active power tends to flow away from such a load [5]. On the other hand, when a transient disturbance event is present in a system, it can often be thought of as an energy sink. For example, in the case of a capacitor switching disturbance, where a bank of discharged capacitors is switched on-line, energy must be supplied to the bank in order to charge the capacitors. Likewise, during a fault, energy is diverted from other loads to the fault path [6]. Therefore, the direction of energy flow through the network during a disturbance is a key indicator of the location of the disturbance source.

The direction of the energy flow is found by examining sampled three-phase voltage and current waveforms provided by recording devices. Using the recorded voltage and current waveforms, the three-phase instantaneous power flow through the recording device is calculated. During steady-state operation, assuming a reasonably balanced system, this power has a fairly constant value. However, when a disturbance event occurs, the network temporarily falls out of steady-state operation, causing a change in the instantaneous power flow.

The difference in the total three-phase instantaneous power and the steady-state three-phase instantaneous power is defined as the "disturbance power" (DP). Since the steady-state instantaneous power is fairly constant, the DP is

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approximately zero except when the disturbance is on. Therefore, a nonzero value for DP indicates the change in the instantaneous power flow caused by the disturbance event. The integral of the DP, the "disturbance energy" (DE), likewise represents the change in energy flow through the recording device due to the disturbance, since the DP makes little contribution to this integral when no disturbance is present, being approximately zero. Information about changes in the DP and DE allow us to make a decision about the location of the disturbance, as energy tends to flow towards the disturbance source. The next section uses a simple analytical example to illustrate the location procedure.

III. PROCEDURE

To illustrate the use of the DP and DE to locate the disturbance sources, a simple analytical fault example is presented. Fig. 1 shows the single-phase network that is used in this section.

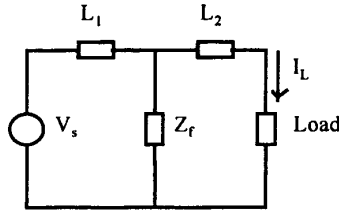


Fig. 1. Circuit Diagram for the Analytical Fault Example.

The circuit consists of a sinusoidal voltage source $V_s = 7.62 \text{ kV}_{\text{RMS}}$, two line segments represented by the inductances L_1 and L_2 (3.7 and 2.7 mH, respectively), and a 500 HP load at $\text{pf} = 0.8$ lagging, which is modelled with a resistor and inductor connected in parallel. To simulate the presence of a fault, the fault impedance Z_f is connected from the midpoint of the line segments to ground.

Solving for the pre-fault condition (i.e., $Z_f = \infty$), we find that $V_{\text{load}} = 7.53 \text{ kV}_{\text{RMS}}$, and the instantaneous power delivered to the load under normal conditions is given by

$$p(t)_{ss} = 364.3 - 355.7 \cos(2\omega t) - 284.3 \sin(2\omega t) \text{ kW} . \quad (1)$$

To simulate a fault where the load voltage drops to approximately 80% of its nominal value, let $Z_f = 1.9 \Omega$. Assuming the circuit immediately enters into a new steady-state operation, the voltage across the load drops to $6.06 \text{ kV}_{\text{RMS}}$, and the instantaneous power delivered to the load during the fault is

$$p(t)_f = 235.97 - 103.2 \cos(2\omega t) - 276.3 \sin(2\omega t) \text{ kW} . \quad (2)$$

The disturbance power to the load is calculated as

$$DP(t) = p(t)_f - p(t)_{ss} , \quad (3)$$

where $p(t)_f$ and $p(t)_{ss}$ are the instantaneous power delivered to the load during the fault and during normal steady-state operation, respectively. Using (1) and (2) in (3), the disturbance power here is found to be

$$DP(t) = -128.33 + 458.9 \cos(2\omega t) + 7.98 \sin(2\omega t) \text{ kW} . \quad (4)$$

The disturbance energy is calculated by integrating (4) over the duration of the disturbance. Fig. 2 shows a plot of the disturbance energy for this example.

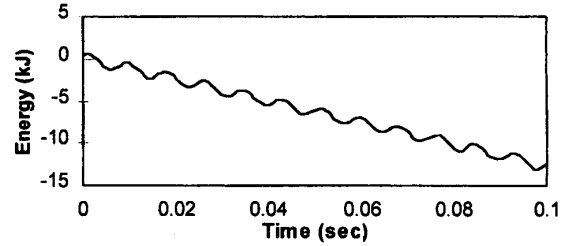


Fig. 2. Plot of Disturbance Energy.

Since this is a single-phase circuit, there is an oscillation present in the disturbance energy, but there is also a general downward trend, indicating a net negative disturbance energy flow through the metering point. Note that this does not represent an *actual* energy flow, as energy is still being delivered to the load even when the fault is on. Rather, this is an indication of the *change* in energy flow, as less power (and energy) is delivered to the load when the fault is on. Since the presence of the fault causes less energy to reach the load, it can be thought of as acting like an energy sink. The direction of the disturbance energy flow away from the load and towards the fault indicates the location of the disturbance source, i.e., the fault impedance Z_f .

IV. EXAMPLES

In this section, we apply the procedure described in Section III to both computer-simulated disturbances and to actual disturbance waveforms for which the disturbance direction is known.

IV.A. ATP Voltage Sag Disturbance

A diagram of the network used for the ATP computer simulation, where locations of recording devices and loads are indicated, is shown in Fig. 3. The arrows indicate the direction of positive power flow for each meter.

First, we simulate a single-phase-to-ground fault located between Meter 1 and Load A that produces a voltage sag throughout the network. The fault has a duration of one cycle, and a fault impedance of 10Ω . The voltage waveform for the faulted phase voltage at Meter 2 is shown in Fig. 4. Fig. 3 shows that the fault is located behind Meter 2, but this

information cannot be obtained from the waveform in Fig. 4 alone. The disturbance energy flow through Meter 2, found by taking the integral of the disturbance power, must be examined to determine the direction to the disturbance source. A plot of the disturbance energy is shown in Fig. 5.

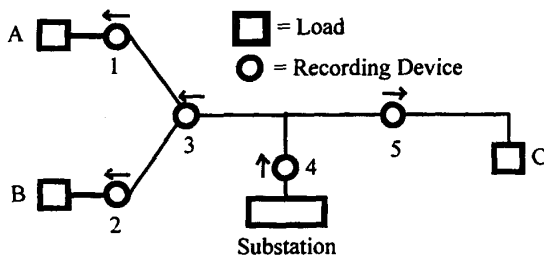


Fig. 3. ATP Distribution Network Diagram.

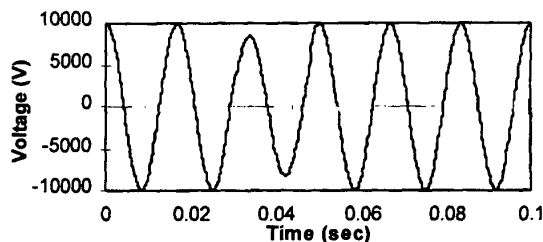


Fig. 4. Faulted Phase Voltage at Meter 2.

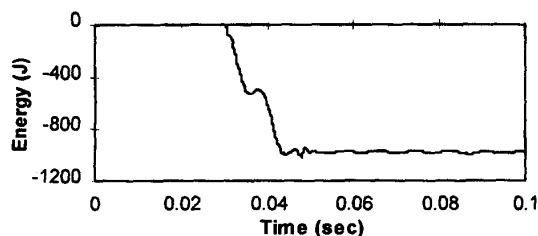


Fig. 5. Disturbance Energy Flow Through Meter 2.

The negative disturbance energy flow in Fig. 5 indicates that the disturbance is located behind Meter 2. While not shown here, the disturbance energy flows through the other meters in the system also indicate the correct directions. By concurrently examining the directions given by all the recording devices, the location of the fault can be accurately pinpointed as being in the line segment in front of Meter 1.

The disturbance power flow through Meter 2, shown in Fig. 6, can be used to confirm the decision made above. Based on our observations of both simulated and recorded disturbances, the polarity of the initial peak of the disturbance power also tends to indicate the direction to the disturbance

source. For example, in Fig. 6, the initial peak is negative, which indicates a disturbance in the negative direction (i.e., behind the meter), which is the same result obtained by looking at the disturbance energy. Since both direction indicators are in agreement, we can place a greater degree of confidence in our decision.

Similar results are also obtained for high and low impedance faults (50Ω and 3Ω , respectively), as well as for balanced three-phase-to-ground faults.

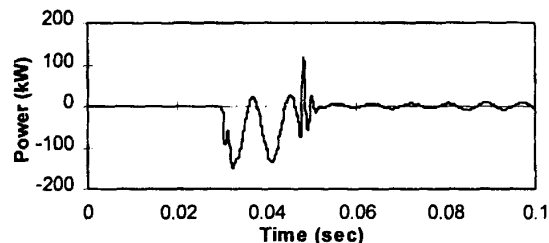


Fig. 6. Disturbance Power for Meter 2.

IV.B. ATP Capacitor Switching Disturbance

Next, a three-phase, 300 kVAr power factor-correction capacitor bank is placed at Load B in Fig. 3. The switches on each capacitor were set to flash over near the peak of the associated phase voltage waveforms. One phase of the recorded voltage at Meter 2 is shown in Fig. 7.

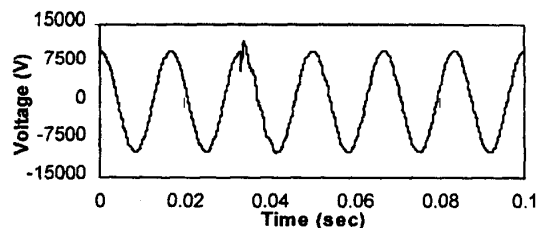


Fig. 7. Cap. Switching Voltage Waveform at Meter 2.

From Fig. 3, the disturbance source in this case is located in front of Meter 2 although the voltage waveform in Fig. 7 seems to give no indication of this. Fig. 8 shows the disturbance energy flow through Meter 2, which settles out to a positive value after the transient disturbance dies out, indicating the correct direction. It is interesting to note that the final value of energy delivered to the capacitor bank given by ATP is approximately equal to the stored energy in the capacitors, given by $\frac{3}{2} C V_{RMS}^2$. In Fig. 9, which shows the disturbance power through Meter 2, the polarity of the initial peak of the disturbance power again indicates the correct direction to the disturbance source.

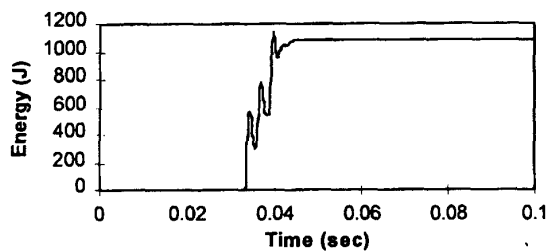


Fig. 8. Disturbance Energy Flow Through Meter 2.

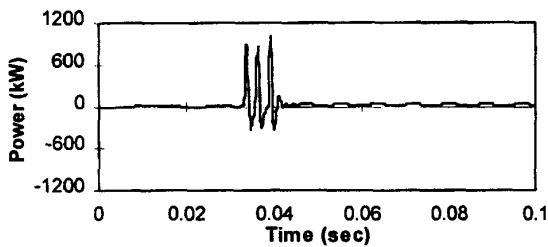


Fig. 9. Disturbance Power for Meter 2.

While the results are not presented here, the other meters on the path from the substation to the capacitor bank at load B (i.e., Meters 3 and 4) also showed a strong positive disturbance energy flow, indicating a disturbance in the forward direction. For the meters off of this path, i.e., Meters 1 and 5, the disturbance energy did not show a significant change. However, since the initial peak of the disturbance power at each meter indicated a backward capacitor switching disturbance, the correct direction for each meter may still be found.

IV.C. Recorded Voltage Sag

Fig. 10 shows the phase A voltage and current for an actual sag disturbance recorded on TU Electric's transmission system near Austin, Texas.

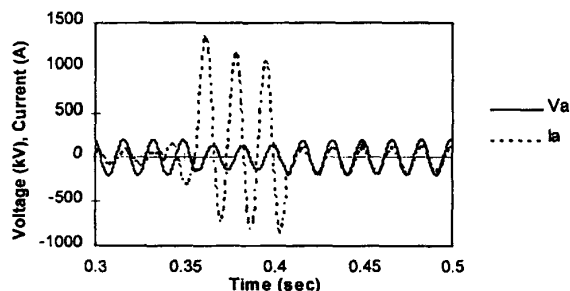


Fig. 10. Voltage and Current for Recorded Voltage Sag.

Before the disturbance event, the current slightly lags the voltage. When the fault is on, however, the current leads the voltage, indicating a fault in the negative direction, or behind the recording device [7]. Fig. 11 shows the disturbance energy through the meter. When the fault clears at approximately 0.4 seconds, the negative value of disturbance energy indicates a fault in the negative direction (i.e., behind the meter), confirming the results given by the phase angle observation. The net energy starts to rise again as the system recovers from the fault, but this is not a concern since it occurs only after the fault has cleared and the disturbance is over.

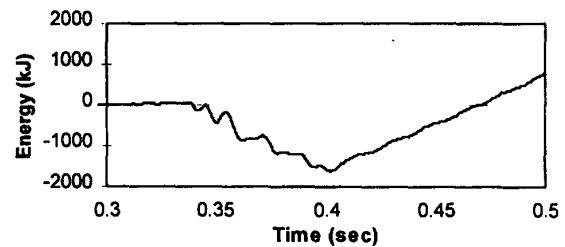


Figure 11. Disturbance Energy, Recorded Voltage Sag.

The disturbance power for this sag disturbance is shown in Fig. 12. As before, the negative initial peak confirms the decision made based on the disturbance energy, and allows the source location to be identified with a greater degree of confidence.

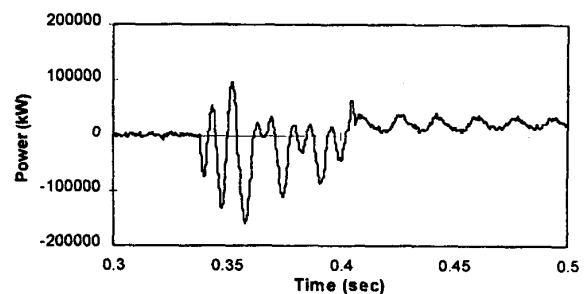


Fig. 12. Disturbance Power, Recorded Voltage Sag.

IV.D. Recorded Capacitor Switching

Fig. 13 shows the voltage for one phase of a capacitor switching event recorded on a distribution feeder in the TU Electric system in Dallas, Texas. The disturbance shown in the figure is known to have been caused by a capacitor switching event that took place behind the meter.

The associated disturbance energy, shown in Fig. 14, oscillates about zero until the disturbance is nearly over (about 0.030 sec), at which time it begins to climb due to the circuit entering a new steady-state. Therefore, at the

effective end of the disturbance, the disturbance energy is “small” and the test is inconclusive. We recommend that “small” be defined as less than 80% of the maximum excursion of the waveform (i.e., $0.80 \cdot 2200 \text{ J}$ in this case).

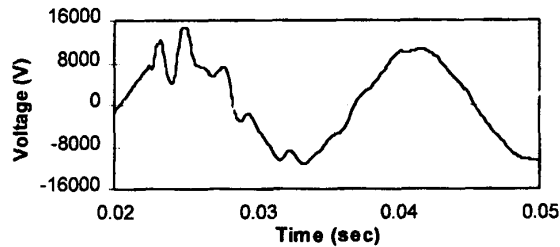


Figure 13. Phase Voltage, Reverse Cap. Switching Event.

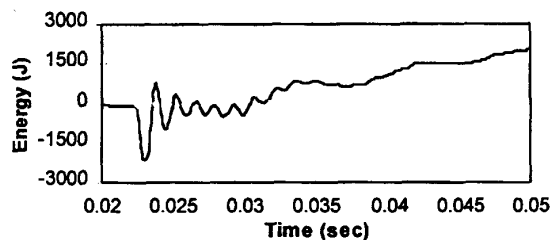


Fig. 14. Disturbance Energy, Reverse Cap. Switching Event.

In a case such as this where the disturbance energy test is inconclusive, the direction may still be determined based on the disturbance power. The initial peak of the disturbance power (Fig. 15) is negative, indicating that the disturbance originates in the negative direction. Thus, even though the energy flow is inconclusive in this case, the initial peak of the disturbance power correctly indicates the direction.

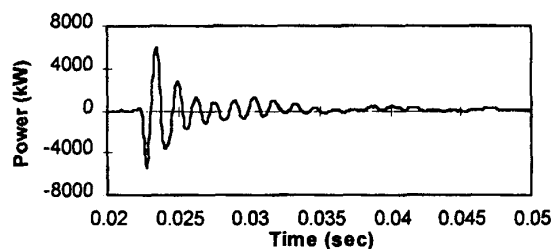


Fig. 15. Disturbance Power, Reverse Cap. Switching Event.

Fig. 16 shows one phase voltage for a capacitor switching event recorded on a TU Electric distribution feeder in Round Rock, Texas. The source of this disturbance is known to have been located in front of the meter. Fig. 17 shows the

associated disturbance energy, while Fig. 18 shows the disturbance power.

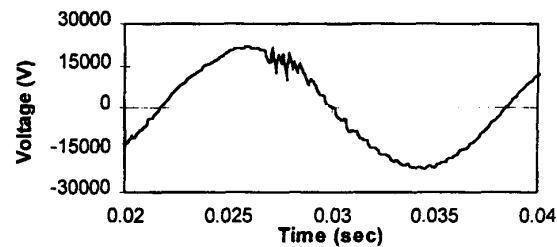


Fig 16. Phase voltage, Forward Cap. Switching Event #F-1.

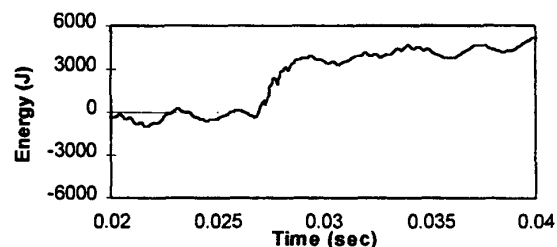


Fig. 17. Disturbance Energy, Forward Cap. Switching Event #F-1.

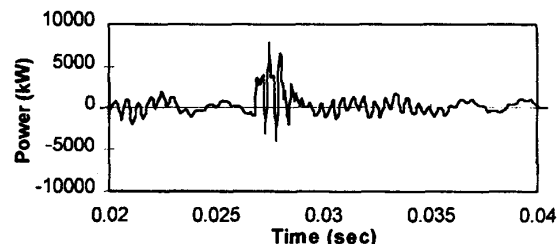


Fig 18. Disturbance Power, Forward Cap. Switching Event #F-1.

As shown in Figs. 17 and 18, both indicators (i.e., disturbance energy and the initial peak of the disturbance power) give the correct direction.

The results for another forward capacitor switching event are shown in Figs. 19, 20, and 21. For this disturbance, the net disturbance energy is positive, while the initial peak of the disturbance power is negative. In this case, the strong positive energy flow indicates that the disturbance must be located in front of the meter (as it actually was). The decision, however, is not confirmed by the initial peak of the disturbance power. Therefore, the confidence level in the decision is not as high as for event F-1, where both indicators give the same result.

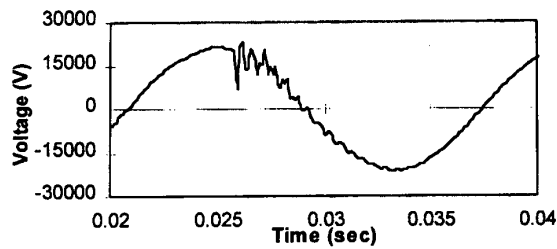


Fig. 19. Phase Voltage, Forward Cap. Switching Event #F-2.

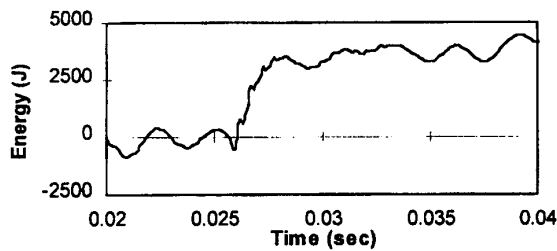


Fig. 20. Disturbance Energy, Forward Cap. Switching Event #F-2.

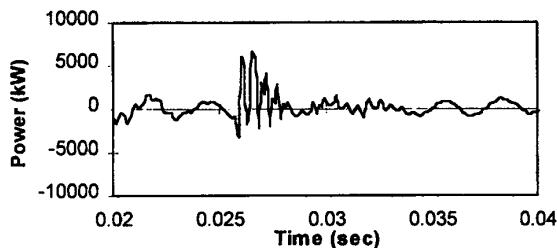


Fig. 21. Disturbance Power, Forward Cap. Switching Event #F-2.

V. CONCLUSIONS

By examining sampled voltage and current waveforms, it is possible to make a judgment as to which side of a recording device a power quality disturbance event originates. This is accomplished by examining the disturbance power and energy flow and the polarity of the initial peak of the disturbance power. If several recording devices are available in a network, the source of the disturbance may be pinpointed with a higher degree of accuracy.

The solution steps are summarized as follows:

- Determine the beginning and end points of the disturbance event.

- Calculate the three-phase instantaneous power (IP) before, during, and after the event.
- Subtract the pre-disturbance steady-state value from IP to find the disturbance power, DP.
- Integrate DP to find the disturbance energy, DE.
- Examine the value of the DE at the end of the disturbance. There are two possible cases:
 - If the final value of DE is greater than or equal to 80% of the peak excursion of DE during the event, then the energy test is conclusive. The disturbance direction is the same as the polarity of the final DE value. If the polarity of the initial peak of the DP matches the polarity of the final DE value, then we have a high degree of confidence. If the polarities do not match, we still declare the disturbance to be in the direction indicated by the DE, but with a lesser degree of confidence.
 - If the final value of the DE is less than 80%, then we say that the energy test is inconclusive. The disturbance direction is determined by examining the polarity of the initial peak of the DP.

While we have shown results for only two major types of disturbances in this paper, the procedure also works well for motor starting disturbances because extra energy is needed to accelerate the motor. We also believe that the methods presented are applicable to other disturbance types as well. Thus, in the future we plan to revise and expand our existing set of rules.

This procedure may prove valuable to utilities interested in locating the sources of recorded disturbances in order to determine, for example, whether a disturbance originated inside or outside of a customer facility.

VI. ACKNOWLEDGEMENTS

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VIII. BIOGRAPHIES

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